WORKED SOLUTION

STEP III 1997, Question 2

Pure • Curve Sketching

STEP III 1997, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q2

The key idea

Taking logarithms turns $x^y = y^x$ into $y \ln x = x \ln y$, and dividing by xy gives $\mathrm{f}(x) = \mathrm{f}(y)$. So the second half is really a question about the first: once we know how many inputs give each output of f, we know how many partners y each x has. The equation always has the obvious solution $y = x$; the interest is in the other one.

Solution

The graph of f

Differentiating with the quotient rule,

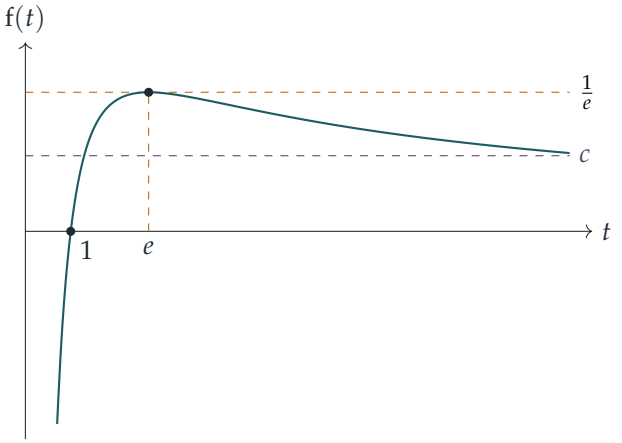
$$\mathrm{f}'(t) = \frac{t \cdot \frac{1}{t} - \ln t}{t^2} = \frac{1 - \ln t}{t^2}, \qquad \mathrm{f}''(t) = \frac{2\ln t - 3}{t^3}.$$

So $\mathrm{f}'(t) > 0$ for $0 < t < e$ and $\mathrm{f}'(t) < 0$ for $t > e$: f is strictly increasing on $(0, e]$ and strictly decreasing on $[e, \infty)$. The maximum value is

$$\mathrm{f}(e) = \frac{1}{e}.$$

Other features for the sketch:

- $\mathrm{f}(t) < 0$ for $0 < t < 1$, $\mathrm{f}(1) = 0$, and $\mathrm{f}(t) > 0$ for $t > 1$.
- As $t \to 0^+$, $\ln t \to -\infty$ and $1/t \to +\infty$, so $\mathrm{f}(t) \to -\infty$: the y-axis is an asymptote.
- As $t \to \infty$, $\mathrm{f}(t) \to 0^+$, since $\ln t$ grows more slowly than t. So the t-axis is an asymptote.
- There is a point of inflection at $t = e^{3/2}$, where $\mathrm{f} = \frac{3}{2}e^{-3/2} \approx 0.33$. Nobody needs it for the counting, but it makes the sketch honest.



How many t give a value c?

Because f is continuous and strictly monotonic on each of $(0, e]$ and $[e, \infty)$, each piece takes any value it reaches exactly once. On $(0, e]$, f increases from $-\infty$ to $\frac{1}{e}$; on $[e, \infty)$ it decreases from $\frac{1}{e}$ towards 0, never reaching it. Hence, for a given value c:

- $c > \frac{1}{e}$: **no** values of t;
- $c = \frac{1}{e}$: **one** value, $t = e$;
- $0 < c < \frac{1}{e}$: **two** values, one in $(1, e)$ and one in (e, ∞) (the horizontal line c in the sketch);
- $c \leq 0$: **one** value, in $(0, 1]$, because the decreasing branch stays strictly positive.

Counting solutions of $x^y = y^x$

For $x, y > 0$ both sides are positive, so we may take logarithms; ln is one-to-one, so nothing is gained or lost:

$$x^y = y^x \iff y \ln x = x \ln y \iff \frac{\ln x}{x} = \frac{\ln y}{y} \iff \mathrm{f}(y) = \mathrm{f}(x),$$

where we divided by $xy > 0$. So, for fixed x, the number of positive y is the number of t with $\mathrm{f}(t) = c$ where $c = \mathrm{f}(x)$. One of them is always $y = x$.

? **Ask yourself** Which of the four cases above can $c = \mathrm{f}(x)$ actually fall into?

Since c is a value of f, the case $c > \frac{1}{e}$ cannot happen. Translating the other cases back into conditions on x:

- $0 < x \leq 1$: then $c = \mathrm{f}(x) \leq 0$, so there is exactly **one** y, namely $y = x$.
- $x = e$: then $c = \frac{1}{e}$, so there is exactly **one** y, namely $y = e$.

- $1 < x < e$ or $x > e$: then $0 < c < \frac{1}{e}$, so there are exactly **two** values of y: $y = x$ and one other. If $1 < x < e$ the other lies in (e, ∞); if $x > e$ it lies in $(1, e)$.

For example, $x = 2$ has partner $y = 4$, since $2^4 = 4^2 = 16$.

Sketching the solution set

The set consists of the line $y = x$ (for all $x > 0$) together with a second curve made of the "other" solutions, which lives entirely in $x > 1$, $y > 1$. Its shape:

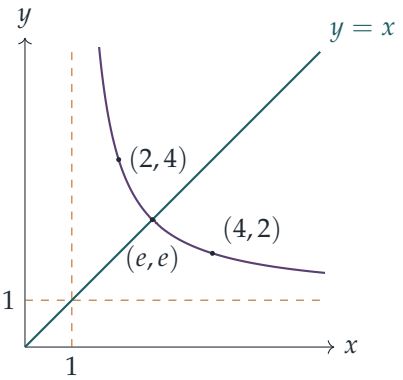
- *Symmetry.* The equation is unchanged when x and y are swapped, so the curve is symmetric in the line $y = x$.
- *Decreasing.* As x increases through $(1, e)$, $\mathrm{f}(x)$ increases, so the partner $y > e$ must move to where f is larger, i.e. y decreases. Similarly for $x > e$. So y is a decreasing function of x on this branch.
- *Asymptotes.* As $x \to 1^+$, $\mathrm{f}(x) \to 0^+$, so the partner $y \to \infty$: the line $x = 1$ is an asymptote. By symmetry, as $x \to \infty$, $y \to 1^+$, and $y = 1$ is an asymptote.
- *Crossing the line $y = x$.* As $x \to e$, the partner also tends to e, so the curve passes through (e, e) and crosses the line there. Near $t = e$, Taylor's theorem with $\mathrm{f}'(e) = 0$ and $\mathrm{f}''(e) = -e^{-3}$ gives $\mathrm{f}(t) \approx \frac{1}{e} - \frac{(t-e)^2}{2e^3}$. So $\mathrm{f}(x) = \mathrm{f}(y)$ forces $(y - e)^2 \approx (x - e)^2$, i.e. $y - e \approx \pm(x - e)$. The plus sign is the line $y = x$. The minus sign is our curve, which therefore crosses the line at right angles, with gradient -1.

? **Ask yourself** Can we write the curve down explicitly?

Yes, if we parametrise by the ratio. Put $y = tx$ with $t > 0$, $t \neq 1$. Then $x^{tx} = (tx)^x$; taking x-th roots gives $x^t = tx$, so $x^{t-1} = t$, and

$$x = t^{1/(t-1)}, \qquad y = t^{t/(t-1)}.$$

At $t = 2$ this gives $(2, 4)$, and at $t = \frac{1}{2}$ it gives $(4, 2)$. As $t \to 1$, both coordinates tend to e, because $t^{1/(t-1)} \to e$. This is how the curve below was drawn.



Checks

- $2^4 = 4^2$ is the famous integer example, and it sits on the curve, in $1 < x < e$ and $x > e$ as predicted.
- Numerically, a fine grid search puts the maximum of f at $t \approx 2.718$ with value $0.367879 \approx 1/e$. Bisection gives the partner of $x = 1.01$ as $y \approx 659$, and of $x = 100$ as $y \approx 1.050$. These are consistent with the asymptotes $x = 1$ and $y = 1$.
- A central difference at $x = e$ gives gradient -1.0000 for the second branch.

Common slips

- Saying every $x \neq e$ has two partners. For $0 < x \leq 1$ the value $\mathrm{f}(x)$ is not positive, and only $y = x$ works.
- Forgetting the line $y = x$ in the final sketch, or drawing the other branch so that it touches the line at (e, e) instead of crossing it.
- Dividing by $\ln x$ or $\ln y$ (which can be zero) instead of by xy.

Beyond STEP

The second curve is a level set of $\mathrm{f}(x) - \mathrm{f}(y)$, and (e, e) is a point where the gradient of that function vanishes. That is why two branches cross there instead of one smooth curve passing through. The implicit function theorem, met in first-year analysis, explains when a level set is locally a graph and what happens when it isn't. Abbott's *Understanding Analysis* is a friendly place to build the foundations. The equation $x^y = y^x$ itself has a long history; Siklos's *Advanced Problems in Mathematics* has more questions in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.